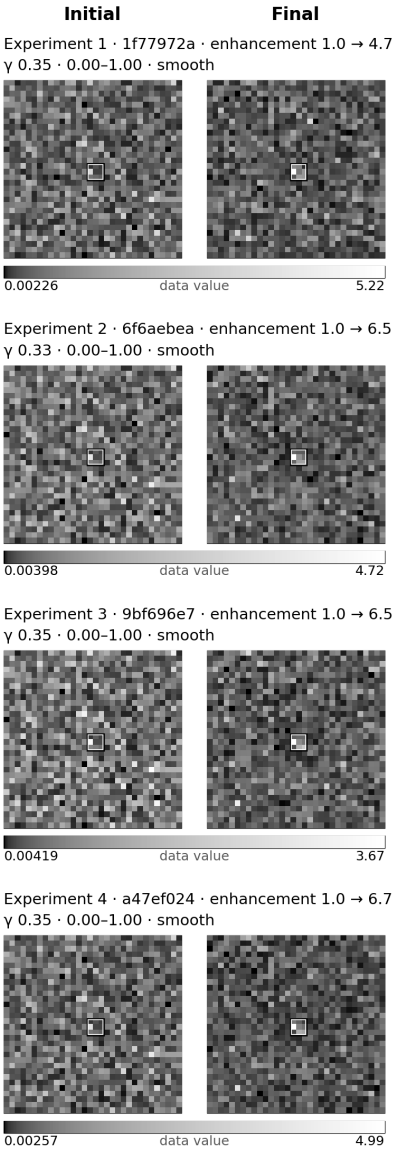
Iteration budget and optical focusing

Research question: How does search iteration budget affect validated focusing when other settings are fixed?

This is a fair budget comparison: the simulated bench, seed, 2×2 macropixels and measurement regions are fixed; only iterations change. Validated enhancement rose from 4.74 to 6.70. The gain was 1.75 from 100 to 300 iterations, then 0.04 and 0.17. Enhancement is focus mean divided by the equally weighted mean of four background-region means.



Initial (left) and Final (right) heatmaps; rows follow the frozen budget order shown below. Grayscale tuning changes presentation, not the underlying measurements. The original raster labels may be small at print size.

Recorded comparison

Budget (iterations)	Final enhancement	Search cost: camera shots / measurement	Validation captures
100	4.74	3 shots	5 fresh
300	6.49	3 shots	5 fresh
600	6.53	3 shots	5 fresh
1000	6.70	3 shots	5 fresh

Verdict: 1000 iterations gave the highest recorded enhancement, but returns diminished after 300. Cost is reported as iterations and 3 camera shots per search measurement (5 fresh validation captures); total shots and runtime were not recorded. These four runs share one search seed, so they do not establish robustness. Next, repeat the highest-budget setting with search_seed=43, changing only the seed. This was a simulation; physical print quality was not validated.